SRC/IRB Approval Signature and Date
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ALAMEDA COUNTY SCIENCE & ENGINEERING FAIR

ACSEF Required Research/Project Plan for General Application
Students Participation in the Science Fair

Link to INSTRUCTIONS: (for extra information,PLEASE READ!!!)

To type into this form, first, make a copy of this Google Document.
This document is VIEW ONLY.
Research Plan Requirements: for all types of projects: The typing area sections will increase in size as you type your answers.
Need to make changes: see Fair ***Link to INSTRUCTIONS***

Your name(s) including each team member if applicable:
⇨ Vincent Lu

Date **(must be dated before data collection starts started)**
⇨ 11/28/2024

Title of your project (10 words or less): ***How do I select an appropriate title***
⇨ Exploring Innovative Machine Learning Models for Soil CO_2 Emission Forecasting

Check all that apply:

This project is a: Science Project _______ Engineering Project _x___

Check one of the following: (note if you check any of the items below you will need to fill out a part of Section 2)

- **Will use Chemicals/Hazardous Devices**
- **Will use Human Participants**
  - **Will use photos and videos of participants for data**
  - **Will not use photos and videos of participants**
- **Will use Vertebrate Animals or Tissues**
- **Will use Potential Harmful Microbes**

N/A

SECTION 1

1. Rationale: include a brief summary of the background that supports your research problem and explain why this research (science or engineering) is important (to humans, and what societal impact will your project address. (See ***Link to INSTRUCTIONS*** for extra information)

In recent years, an increasing trend in CO_2 emissions has been impacting the pace of global warming. The main factor responsible for these emissions is a process called soil respiration in which CO_2 is released from earth's surface to the atmosphere due to decomposition of soil organic matter and plant litter by soil microbes, plant roots, and soil animals. However, accurately predicting CO_2 emissions from soil respiration remains a significant challenge, and the influence of various environmental factors is not yet fully understood.

This research will aim to develop machine learning models to predict CO_2 emissions from soil respiration and identify key driving factors. Additionally, SHAP (SHapley Additive exPlanations) analysis will be applied to enhance model explainability, quantifying the influence of individual features on model outcomes, both at the overall level and for specific predictions.

2. Research Question(s), Hypothesis(es) or Engineering goal(s), expected outcomes: How is this question/hypothesis/goals/outcomes based on your rationale above (see #1)? (See ***Link to INSTRUCTIONS*** for extra information)

- What are the environmental factors in driving soil respiration?
  There are many factors influencing soil respiration, such as temperature, precipitation, and location. Which factors are more important than others?

- Can CO_2 emissions from soil respiration be predicted by observed environmental variables?
  Most of the previous works follow fundamental approaches that simulate the soil respiration process, which is time consuming and computationally expensive. Can soil respiration be forecasted by observable environmental factors without modeling the complex underlying process?

- Is there an effective and robust way to quantify the influence of individual features on model outcomes, both at the overall level and for specific predictions?
  As modern machine learning models are usually viewed as black-box models, can I explain how the models make predictions? What factors have strong predictive power?

3. Materials, Procedures (experimental design, diagrams/schematics (expected of engineering projects), what data you are collecting, what units of measure are being used) (See ***Link to INSTRUCTIONS*** for extra information)

a. **Materials being used: (include everything including amount, source of microbes or chemicals)**

Model development:

- Python Software Foundation. (n.d.). *Python*. Retrieved from https://www.python.org
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., ... & Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. Retrieved from https://scikit-learn.org/stable/
- TensorFlow Team. (n.d.). *TensorFlow and Keras: A guide to Keras*. Retrieved from https://www.tensorflow.org/guide/keras
- Lundberg, S. M., & Lee, S. I. (n.d.). *SHapley Additive exPlanations (SHAP)*. Retrieved from https://shap.readthedocs.io/en/latest/
- Jian, J., Vargas, R., Anderson-Teixeira, K. J., Stell, E., Herrmann, V., Horn, M., Kholod, N., Manzon, J., Marchesi, R., Paredes, D., & Bond-Lamberty, B. (2021). A restructured and updated global soil respiration database (SRDB-V5). Earth System Science Data, 13(1), 255–267. Retrieved from https://github.com/bpbond/srdb

Demonstration: soil samples, water, plastic tubes, containers, Color Charts for reading soil and compost results, pH test strips, soil detector probes. (For demonstration only)

b. **Experimental Design:**.(include drawings if appropriate) NOTE: any engineering or science project that involves building a device MUST have labeled diagrams)

My research can be divided into 4 phases:

Phase 1: Project preparation

- Literature Review
- Identify research gap
- Propose solution
- Search for Database

Phase 2: ML Model Development

- Data preparation
- Data exploratory analysis
- Feature selection

- Model architecture
- Model training
- Model evaluation

Phase 3: Model explainability and Deep dive
- Model comparison
- Feature importance analysis at the overall level
- Impact of features on individual predictions
- Interaction among features

Phase 4: Application and demonstration
- Demonstrate how input variables impact soil respiration

c. **Risk and Safety:** Identify any potential risks and safety precautions, be specific. (and complete Section 2 below the bibliography if you have any safety/risks issues). If none, state why there are no safety or risks…e.g. This project was computer based). (See ***Link to INSTRUCTIONS*** for extra information)

There are no risks because the data is open-sourced and the project is computer-based.

d. **Data Analysis** (describe what procedures you will use to analyze your data/results:(See ***Link to INSTRUCTIONS*** for extra information)

The SRDB-V5 dataset includes data collected from previously published research. The fields include the annual CO_2 emissions, spatial/temporal, climatic, land use, soil composition, and other information. Selected key fields are listed in Table 1 below.

The procedure of data analysis includes the following:
- Data cleaning: excluding records that are marked with issues.
- Examine each field, such as distribution
- Exclude outliers
- Conduct correlation analysis among different factors
- Select the target variable that best represents CO_2 emission from soil respiration.

Once data preparation is done, we will proceed to develop machine learning models with the clean dataset. The steps include
- Explore XGBoost and Deep Neural Network models to predict soil respiration.
  - Feature selection
  - Model architecture selection

- Hyper-parameter optimization
  For XGBoost, parameters such as learning rate, number of estimators, maximum tree depth, and subsampling rate will be optimized.
  For the DNN model, the number of layers, the number of neurons in the dense layers, and learning rate will be.
- Conduct SHAP analysis to analyze the impact of different input variables.

Table 1: Key Fields in SRDB Dataset

Category	Field Name	Definition
Rs_annual	Rs_annual	Annual CO2 flux from soil respiration
Spatial/temporal	Study_year	Year study was performed (middle year if multiple years)
	YearsOfData	Years of data in the study
	Latitude	Latitude of the site
	Longitude	Longitude of the site
	Biome	Biome type
	Elevation	Elevation of the site
Climate	MAT	Reported mean annual temperature
	MAP	Reported mean annual precipitation
Land use	Ecosystem_type	Ecosystem type (grassland, forest, etc)
	Ecosystem_state	Ecosystem state (managed, unmanaged, natural)
Soil	Soil_type	Soil description (classification and texture)
	Soil_drainage	Soil drainage (dry, wet)
	Soil_BD	Soil bulk density
	Soil_CN	Soil C:N ratio
	Soil_sand	Sand ratio
	Soil_silt	Silt ratio
	Soil_clay	Caly ratio
Vegetation	Leaf_habit	Dominant leaf habit (deciduous, evergreen)
	Stage	Developmental stage (aggrading, mature)
	LAI	Leaf area index at site
Carbon	BA	Basal area at site
	C_veg_total	Total carbon in vegetation at site
	C_AG	Total carbon in aboveground vegetation

	C_BG	Total carbon in belowground vegetation
	C_CR	Total carbon in coarse roots
	C_FR	Total carbon in fine roots
	C_soilmineral	Total carbon in soil organic matter
	C_soildepth	Depth to which soil C recorded

Use machine learning models such as XGBoost and Deep Neural Network models to predict soil respiration and use SHAP analysis to analyze the impact of different input variables.

4. Bibliography: Create a numbered list (**not a Numbered paragraph**) of at minimum **FIVE** major references (e.g. science journal articles, books, internet sites, acknowledgment of the use of primary data sources; open or with permission).

- **All WEB LINKS MUST BE INACTIVE. How to remove Active Web Links(methods) Click this link to see the methods; Please fix them before you upload your research plan to zFairs.**
- **Please use the APA style on MyBib to generate your references;**

1. Lawrence, D. M., Fisher, R. A., Koven, C. D., Oleson, K. W., Swenson, S. C., Bonan, G., ... & Wieder, W. R. (2019). The Community Land Model version 5: Description of new features, benchmarking, and impact of forcing uncertainty. Journal of Advances in Modeling Earth Systems, 11(12), 4245–4287.
2. Lienert, S., & Joos, F. (2018). A Bayesian ensemble data assimilation to constrain model parameters and land-use carbon emissions. Biogeosciences, 15(9), 2909–2930.
3. Ito, A. (2019). Disequilibrium of terrestrial ecosystem CO_2 budget caused by disturbance-induced emissions and non-CO_2 carbon export flows: A global model assessment. Earth System Dynamics, 10(4), 685–709.
4. Adachi, M., Ito, A., Uchida, M., Matsuura, Y., Yonemura, S., ... & Hararuk, O. (2017). Estimation of global soil respiration by accounting for land-use changes derived from remote sensing data. *Journal of Environmental Management, 200*, 97–104.
5. Bahn, M., Reichstein, M., Davidson, E. A., Grüneberg, E., Jung, M., ... & Janssens, I. A. (2010). Soil respiration at mean annual temperature predicts annual total across vegetation types and biomes. *Biogeosciences, 7*(7), 2147–2157.

6. Jian, J., Steele, M. K., Day, S. D., & Thomas, R. Q. (2020). Prediction of annual soil respiration from its flux at mean annual temperature. *Agricultural and Forest Meteorology, 287*, 107961.
7. Liu, J., Hu, J., Liu, H., et al. (2024). Global soil respiration estimation based on ecological big data and machine learning model. *Scientific Reports, 14*, 13231.
8. Lu, H., Li, S., Ma, M., et al. (2021). Comparing machine learning-derived global estimates of soil respiration and its components with those from terrestrial ecosystem models. *Environmental Research Letters, 16*(5), Article 054048.

5. Required Statement of AI involvement:

Please note the following update to our 2024 rules: AI use and information (from Regeneron ISEF/Society for Science) International Science Fair Rules:

This statement clarifies the expectation that students will not have used an AI tool to provide information in any forms, documents, and materials such as directly providing text for writing that they are indicating it is their own. Instead, if students used AI they should cite the use of AI as they would other sources of materials used in the ideation and implementation of their project or application forms. Failure to follow this rule will be considered plagiarism and projects will automatically be disqualified DQ.

ACSEF requires the following statement in the research plan ideation to be answered.

Check ONE statement A or B as being true for your application and fair materials:

A. **No __x__ I/We did not use any form of AI tools to provide any information on application forms, documents, or fair materials.**
B. **Yes ___I/We used AI tools to provide information on application forms, documents or fair materials and will use APA format to cite each source of AI used on each form, document, or other fair materials.**

Use the link below to correctly format your AI citation in APA style - https://libguides.mcmaster.ca/cite-gen-ai/apa See the second section titled "Citing an AI Model in APA

Please save this document as a PDF when you are done to add to the other required application documents for applying to participate in the science fair.

---------------**End of General Requirements for All Types of Projects**--------------------------